



Benchmarking AI Workloads on GPUs

A comprehensive guide for ML engineers and infrastructure teams evaluating GPU performance for AI workloads

Why Benchmark GPUs for AI?

Performance Varies by Task

Not all GPUs perform equally across different AI workloads - architecture matters for specific operations

Informed Decision Making

Benchmarks guide **hardware selection & optimization**, preventing underutilization of costly GPU resources

Deployment Strategy

Helps compare **cloud vs on-prem performance** to optimize total cost of ownership (TCO)

Future Planning

Essential for AI infrastructure planning, scaling roadmaps, and capacity forecasting

Types of GPU Benchmarks



Synthetic

Raw FLOPs, memory bandwidth (Geekbench, CUDA samples)

AI Framework

TensorFlow/PyTorch training speed on standard datasets

Model-Specific

ResNet, BERT, GPT benchmarks with real architecture

Inference

Latency & throughput tests under production conditions

A comprehensive benchmarking strategy combines all four types to build a complete performance profile

Key Metrics to Track

32T

FLOPs

Floating point operations per second - raw computational power

1250

Throughput

Samples/sec or tokens/sec processed during training or inference

15ms

Latency

Response time per inference - critical for real-time applications

24GB

Memory

VRAM utilization per batch - often the limiting factor for large models

3.5x

Energy Efficiency

Performance/watt ratio - particularly important for LLM training and HPC

Effective benchmarking tracks all metrics across different model sizes and batch configurations

Benchmarking Tools



MLPerf

Industry standard benchmark suite for training & inference with peer-reviewed methodologies



NVIDIA Nsight Systems

Low-level profiling at kernel level with detailed execution timeline analysis



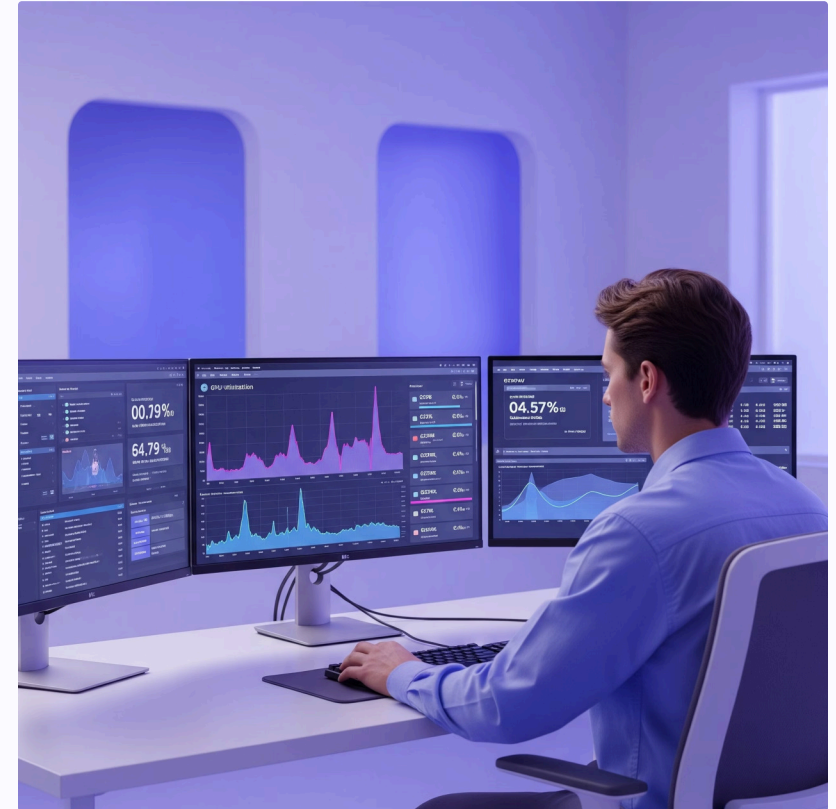
PyTorch/TensorFlow Profilers

Framework-specific tools for detailed operation timing and memory tracking



nvidia-smi

Basic but essential tool for live utilization monitoring and thermal tracking



Benchmarking Training Workloads

Choose Representative Models

ResNet-50, BERT, GPT-2 provide diverse architecture profiles (CNN, Transformer)

Measure Key Metrics

Time per epoch, samples/sec across different batch sizes and precision formats

Test Precision Impact

Compare FP32 vs FP16/BF16 precision to evaluate Tensor Core utilization

Evaluate Scaling

Test **scaling efficiency** (1 GPU vs multi-GPU) to identify communication bottlenecks

Effective training benchmarks detect whether your workload is memory-bound or compute-bound





Benchmarking Inference Workloads

Key Inference Metrics

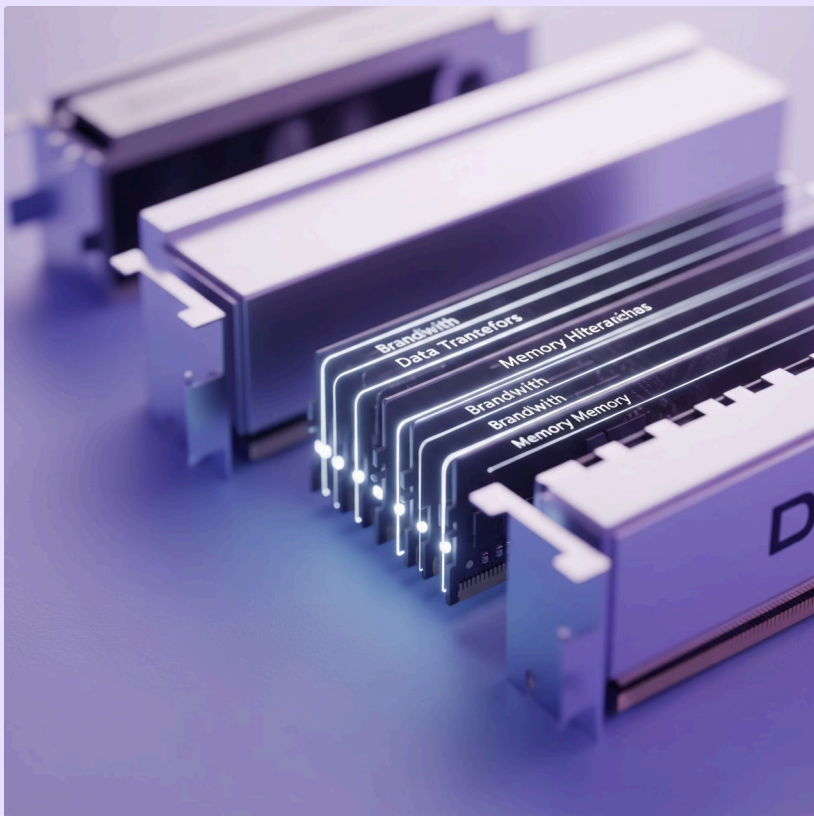
- 1 **Latency per request** - p50, p95, p99 percentiles under various loads
- 2 **Throughput (QPS)** - queries per second at maximum sustainable load
- 3 **Batch size optimization** - finding optimal response time vs throughput

Optimization Tools

- NVIDIA [Triton Inference Server](#)
- TensorRT for CUDA optimization
- ONNX Runtime for cross-platform deployment
- DeepSpeed for large model inference
- Tensor parallelism for LLMs

📌 Inference optimization is critical for production deployments where latency impacts user experience and cost

GPU Memory Benchmarks



Bandwidth Tests

Measure copy speeds: **host** ↔ **device** ↔ **device** using CUDA bandwidthTest

VRAM Capacity

Determine maximum batch size and model size before OOM errors

Memory Fragmentation

Identify wasted GPU space during long-running training jobs

Profiling Tools

PyTorch memory profiler, Nsight Compute memory analysis

Memory is often the true bottleneck in LLM training and high-resolution computer vision

Benchmarking Multi-GPU Scaling



Linear vs. Actual Scaling

Compare **theoretical linear scaling** vs. real-world performance with communication overhead



Communication Fabric

Test PCIe vs NVLink vs InfiniBand topologies using NCCL AllReduce benchmarks



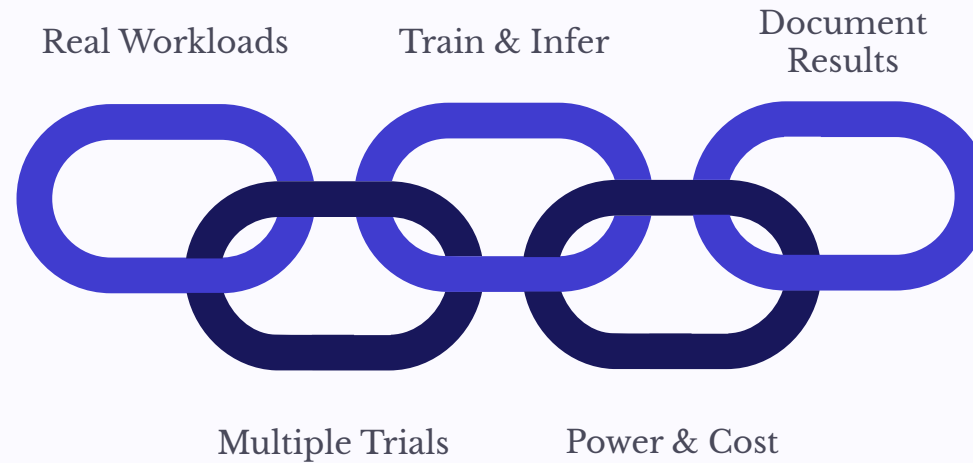
Distributed Training

Benchmark **DDP (DistributedDataParallel)** and model parallelism efficiencies

Multi-GPU scaling reveals the true efficiency of your infrastructure at scale and helps identify communication bottlenecks



Best Practices for GPU Benchmarks



Pro Tip: Establish a regular benchmarking cadence as part of your MLOps process. Re-benchmark when updating frameworks, drivers, or model architectures.

Standardize your benchmarking methodology to ensure consistent, reproducible results across teams and projects



Key Takeaways

Strategic Value

Benchmarking ensures **right GPU choice & efficiency** for your specific AI workloads

Comprehensive Metrics

Measure FLOPs, throughput, latency, and memory usage across diverse scenarios

Professional Tools

Leverage MLPerf, Nsight, and AI framework profilers for detailed insights

Benchmarking is the science of matching AI workloads to hardware

Implementing a rigorous benchmarking strategy will:

- Reduce infrastructure costs
- Improve model training time
- Optimize inference latency
- Support informed scaling decisions